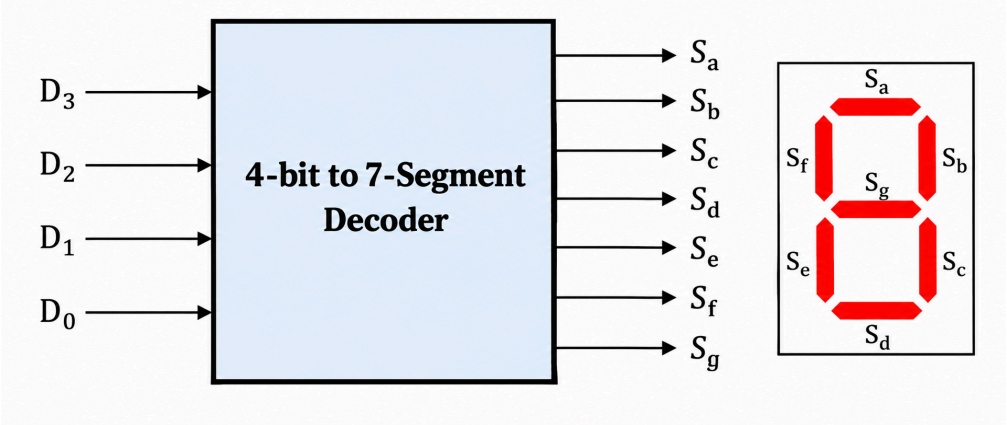
M YASHAS
PES1UG25EC138
CADD ASSIGNMENT 1

Problem statement : Implement optimized seven segment display decoder using xilinx vivado

Black-Box Structure



Truth Table

Truth Table for 4-bit to 7-Segment Decoder												
Decimal Number	Inputs				Outputs (7-Segment)							
	D_3	D_2	D_1	D_0	S_a	S_b	S_c	S_d	S_e	S_f	S_g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1
10	1	0	1	0	X	X	X	X	X	X	X
11	1	0	1	1	X	X	X	X	X	X	X
12	1	1	0	0	X	X	X	X	X	X	X
13	1	1	0	1	X	X	X	X	X	X	X
14	1	1	1	0	X	X	X	X	X	X	X
15	1	1	1	1	X	X	X	X	X	X	X

Kmap Simplifications

KMAP :-

i) S_a

D_3D_2 \ D_1D_0	00	01	11	10
00	1		1	1
01		1	1	1
11	X	X	X	X
10	1	1	X	X

$$S_a = D_3 + D_1 + D_2D_0 + \bar{D}_2\bar{D}_0$$

ii) S_c

1	1	1	
1	1	1	1
X	X	X	X
1	1	X	X

$$S_c = D_2 + \bar{D}_1 + D_0$$

iii) S_b

D_3D_2 \ D_1D_0				
	1	1	1	1
1			1	
X	X	X	X	X
1	1	X	X	

$$S_b = \bar{D}_2 + D_1D_0 + \bar{D}_1\bar{D}_0$$

iv) S_d

D_3D_2 \ D_1D_0				
1		1	1	
	1		1	
X	X	X	X	
1	1	X	X	

$$S_d = \bar{D}_2\bar{D}_0 + D_1\bar{D}_0 + D_2\bar{D}_1D_0 + \bar{D}_2D_1$$

Date :

v) Se

D_3D_2 \ D_1D_0	00	01	11	10
00	1			1
01				1
11	X	X	X	X
10	1		X	X

$$S_e = \overline{D_0}(\overline{D_2} + D_1)$$

vi) SF.

D_3D_2 \ D_1D_0	00	01	11	10
00	1			
01	1	1		1
11	X	X	X	X
10	1	1	X	X

$$S_F = D_3 + D_2\overline{D_1} + D_2\overline{D_0} + \overline{D_1}\overline{D_0}$$

vii) SG.

D_3D_2 \ D_1D_0	00	01	11	10
00			1	1
01	1	1		1
11	X	X	X	X
10	1	1	X	X

$$S_G = D_3 + D_2\overline{D_1} + \overline{D_2}D_1 + D_1\overline{D_0}$$

Design Code:

```
module seven(
    input [3:0] D,
    output [6:0] S
    );
    assign S[6] = ~( D[3] | D[1] | (D[2] & D[0]) | (~D[2] & ~D[0]) );                                  // Sa
        assign S[5] = ~( ~D[2] | (D[1] & D[0]) | (~D[1] & ~D[0]) );                                     // Sb
        assign S[4] = ~( D[2] | ~D[1] | D[0] );                                                         // Sc
        assign S[3] = ~( (~D[2] & ~D[0]) | (D[1] & ~D[0]) | (D[2] & ~D[1] & D[0]) | (~D[2] & D[1]) );// Sd
        assign S[2] = ~( ~D[0] & (~D[2] | D[1]) );                                                      // Se
        assign S[1] = ~( D[3] | (D[2] & ~D[1]) | (D[2] & ~D[0]) | (~D[1] & ~D[0]) );                   // Sf
        assign S[0] = ~( D[3] | (D[2] & ~D[1]) | (~D[2] & D[1]) | (D[1] & ~D[0]) );
endmodule
```

Testbench Code:

```
`timescale 1ns / 1ps

module seven_segment_decoder_tb;

    logic [3:0] D;
    logic [6:0] S;

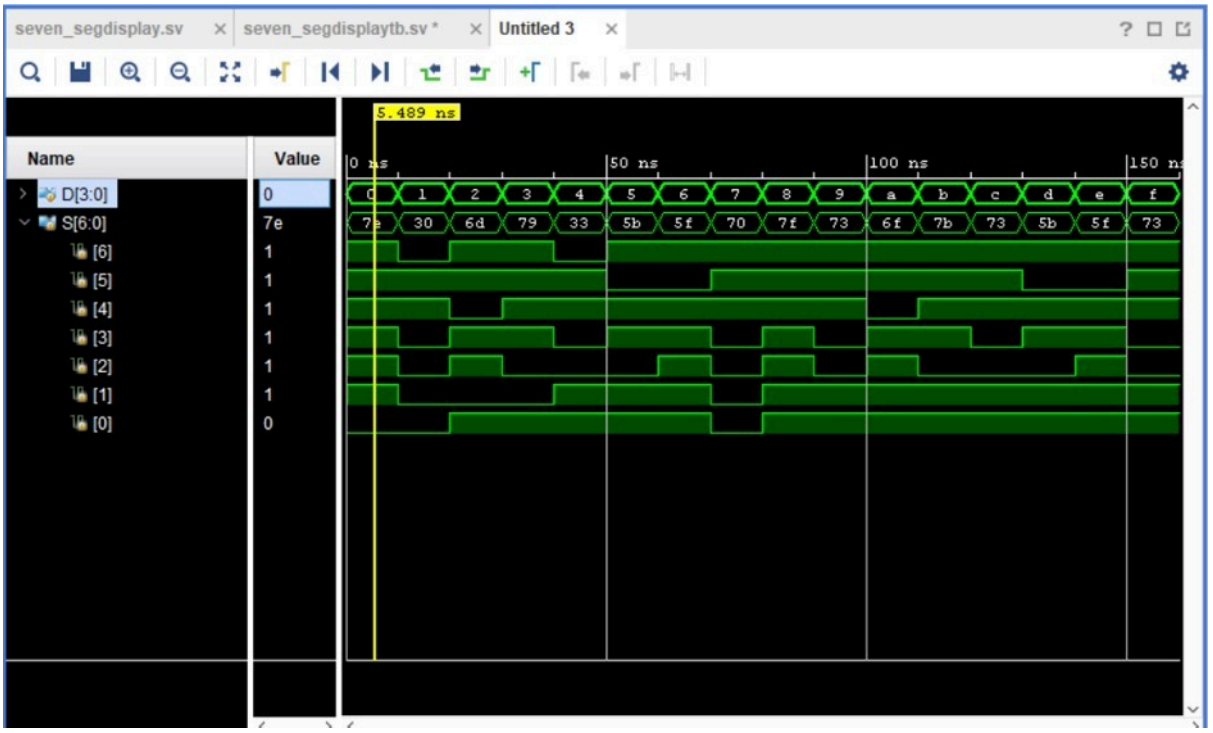
    seven_segment_decoder DUT (
        .D(D),
        .S(S)
    );

    initial begin
        for (int i = 0; i < 16; i++) begin
            D = i;
            #10;
        end

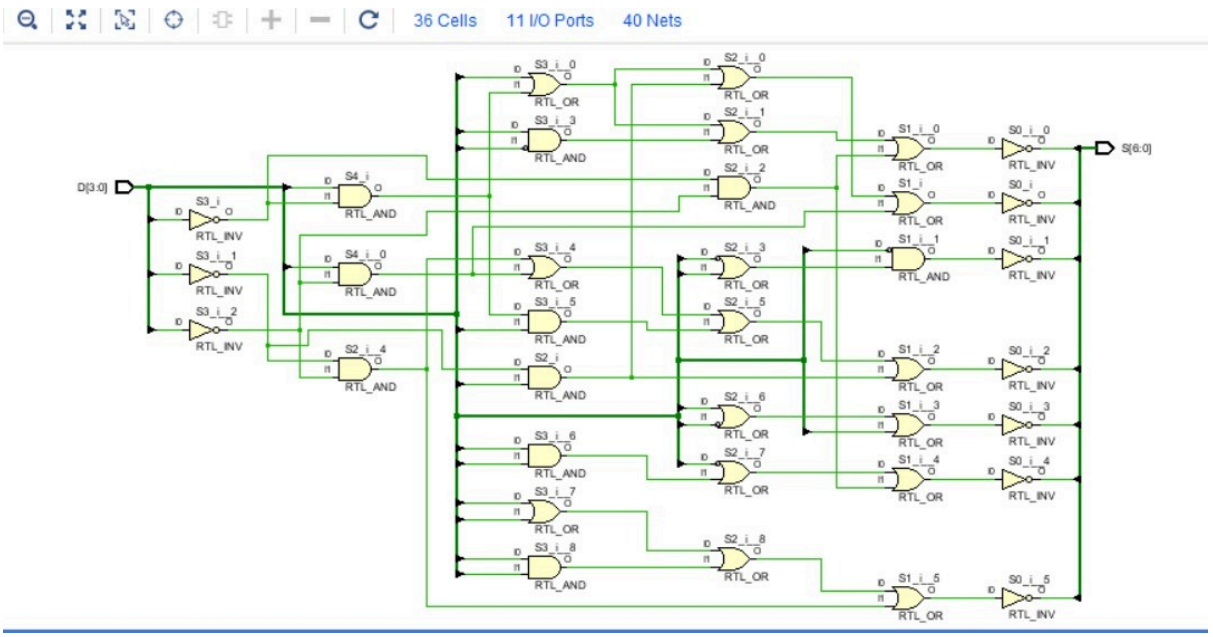
        $finish;
    end

endmodule
```

Simulation Graph:



RTL Schematic :



Conclusion :

The seven segment display decoder was implemented in SystemVerilog and simulated in Xilinx Vivado. The Boolean expressions for the segments Sa to Sg were minimized using K-MAP taking 10-15 as don't cares which will help us to reduce to a small SOP using less number of gates.